Nikita Kiselev

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kisnikser.github.io

Check my Resume (one-page version)

Curriculum Vitæ

SUMMARY

Research Engineer focused on advanced generative diffusion and multimodal models. Experienced in post-training (SFT, RLHF) of image and video generation systems, controllable editing pipelines, and model alignment. Contributor to Kandinsky 5.0 and 4.1. Strong mathematical background in machine learning and optimization with multiple top-tier publications. Lecturer in deep learning and generative models.

Professional Skills

1. **Generative AI:** diffusion models, text-to-video, image-to-video, text-to-image, image-to-image, alignment (SFT, RLHF), controllable generation (ControlNet), adapters (IP-Adapter, Face-Adapter), style transfer, instruction-based image editing, image quality assessment.
2. **Computer Vision:** medical images reconstruction and forecasting (fMRI, EEG).
3. **Optimization:** sample size determination, loss landscape, decentralized optimization.

WORK EXPERIENCE

Kandinsky Lab, Sber AI — **Moscow, Russia**

Research Engineer — *Jan 2025 – Present*

Kandinsky 5.0 (text-to-image, image-to-image):

- Post-Training: SFT, Instruct SFT.
- Alignment: RLHF (DPO, DRAFT, GRPO, DGPO).
- Applications: Text Rendering, Interior Design, Stylization.

Kandinsky 4.1 (text-to-image, text-to-video, image-to-video):

- Performed text-to-image SFT via innovative LoRA-Soup technique, significantly improved image generation quality.
- Text-to-video LoRA fine-tuning for Russian culture code. Trained 70+ LoRA adapters on different characters, distilled them into text-to-video generation of Kandinsky 4.1.
- Image-to-video generation of Kandinsky 4.1. Developed a new architecture and trained the model maintaining structural dynamics.

Research Intern — *Jun 2024 – Dec 2024*

Kandinsky Image Editing:

- Interior Design with Kandinsky 3 ControlNet for Domclick. Fine-tuned ControlNet on the interiors domain, implemented regional prompting for arbitrary segmentation masks and utilized ControlNet image2image editing.
- Kandinsky 3 ControlNet Style (DepthPro) that outperforms ControlNet Union SDXL by 100% on ImageReward. Trained ControlNet over 300k steps on the 3M dataset, used DepthPro condition for the best image preserving and style editing.

Research Center for Artificial Intelligence, Innopolis University — **Innopolis, Russia**

Mathematician-programmer — *Sep 2024 – Jul 2025*

- Research work in the field of optimization

Laboratory of Mathematical Methods of Optimization, MIPT — **Moscow, Russia**

Technician — *Oct 2023 – Apr 2024*

- Research work in the field of optimization
- Decentralized optimization with coupled constraints
- Network design problem

EDUCATION

Moscow Institute of Physics and Technology **Moscow, Russia**
MSc in Computer Science *Sep 2024 – Present*

- Thesis: Sample Size Determination for Parametric Models
- Advisor: Andrey Grabovoy, PhD
- GPA: 5/5

Moscow Institute of Physics and Technology **Moscow, Russia**
BSc in Applied Mathematics and Physics *Sep 2020 – Aug 2024*

- Thesis: Bayesian Sample Size Estimation
- Advisor: Andrey Grabovoy, PhD
- GPA: 4.88/5 (with honours)

PUBLICATIONS

Technical Reports

1. Kandinsky Lab, Sber AI
   Kandinsky 5.0: A Family of Foundation Models for Image and Video Generation
   Top-1 Daily & Weekly & **Monthly Paper** on Hugging Face (November 2025)

Conference Proceedings Papers: 1 × A*, 1 other

1. D. Yarmoshik, A. Rogozin, N. Kiselev, D. Dorin, A. Gasnikov, D. Kovalev
   Decentralized Optimization with Coupled Constraints
   ICLR 2025, **A***
2. V. Meshkov, N. Kiselev, A. Grabovoy
   ConvNets Landscape Convergence: Hessian-Based Analysis of Matricized Networks
   2024 Ivannikov ISPRAS Open Conference (ISPRAS)

Journal Papers: 1 × Q1, 3 × Q2

1. D. Dorin, N. Kiselev, A. Grabovoy, V. Strijov
   Forecasting fMRI Images From Video Sequences: Linear Model Analysis
   Health Information Science and Systems, **Q1**
2. N. Kiselev, A. Grabovoy
   Unraveling the Hessian: A Key to Smooth Convergence in Loss Function Landscapes
   Doklady Mathematics, Q2
3. N. Kiselev, A. Grabovoy
   Sample Size Determination: Likelihood Bootstrapping
   Computational Mathematics and Mathematical Physics, Q2
4. N. Kiselev, A. Grabovoy
   Sample Size Determination: Posterior Distributions Proximity
   Computational Management Science, Q2

Accepted Journal Papers: 1 × VAK K1

1. N. Kiselev, V. Meshkov, A. Grabovoy
   Robust Convergence of Loss Landscapes through Distributional Averaging
   Proceedings of the ISP RAS, VAK K1

Preprints

1. E. Petrov, N. Kiselev, V. Meshkov, A. Grabovoy
   Closing the Curvature Gap: Full Transformer Hessians and Their Implications for Scaling Laws
   Submitted to ICLR 2026, A*

Conference Theses

1. N. Kiselev, V. Meshkov, A. Grabovoy
   Sufficient sample size and its connection to the loss landscape convergence
   Proceedings of the Mathematical Methods for Pattern Recognition 2025 Conference
2. N. Kiselev, A. Grabovoy
   Loss landscape convergence as a sufficient sample size property
   Proceedings of the 67th MIPT All-Russian Scientific Conference
3. V. Meshkov, N. Kiselev, A. Grabovoy
   ConvNets landscape convergence: Hessian-based analysis of matricized networks
   Proceedings of the 67th MIPT All-Russian Scientific Conference
4. N. Kiselev, A. Grabovoy
   Determining a sufficient sample size based on the a posteriori distribution of model parameters
   Proceedings of the 66th MIPT All-Russian Scientific Conference
5. D. Dorin, N. Kiselev, A. Grabovoy
   Spatial and temporal methods of time series analysis
   Proceedings of the 66th MIPT All-Russian Scientific Conference

POSTER SESSIONS

- October 21, 2024
  Spatio-Temporal fMRI Analysis in Visual Stimuli Decoding: Linear Model Forecasting & Voxel Weighing
  Neuroinformatics 2024

TALKS

- December 10, 2025
  Kandinsky: image and video diffusion models fine-tuning for personalization
  Salute, Giga!
- December 10, 2025
  Robust Convergence of Loss Landscapes through Distributional Averaging
  2025 Ivannikov ISPRAS Open Conference (ISPRAS)
- November 20, 2025
  Kandinsky: launching and fine-tuning image and video generation models
  AI Journey 2025
- September 25, 2025
  Sufficient sample size and its connection to the loss landscape convergence
  Mathematical Methods for Pattern Recognition
- April 5, 2025
  Loss landscape convergence as a sufficient sample size property
  67th MIPT All-Russian Scientific Conference
- April 6, 2024
  Determining a sufficient sample size based on the a posteriori distribution of model parameters
  66th MIPT All-Russian Scientific Conference

TEACHING

Diffusion Models — **Standard Data × Sber University**
Lecturer — *Nov 2024 – Present*

- Implementation Details: DDIM, noise schedulers, guidance overexposure, noise offset, multiple text encoders, transformer-based diffusion models, MM-DiT.
- Video Diffusion Models: Deforum, AnimateDiff, Wan.

Deep Learning — **Moscow Institute of Physics and Technology**
Lecturer — *Sep 2024 – Present*

- Optimization, Regularization. (lecture, seminar)
- Initialization, Normalization, CNN. (lecture, seminar)
- Generative Models: Autoregression, VAE, GAN, Diffusion, Flow Matching.
- Acceleration: Quantization, Pruning, Distillation, KV-Cache, Flash Attention.

PROJECTS

- **HippoTrainer: Gradient-Based Hyperparameter Optimization**
  A Python library for gradient-based hyperparameter optimization, implementing cutting-edge algorithms that leverage automatic differentiation to efficiently tune hyperparameters.
- **Just Relax It: Discrete Variables Relaxation**
  A cutting-edge Python library designed to streamline the optimization of discrete probability distributions in neural networks, offering a suite of advanced relaxation techniques compatible with PyTorch.
- **Models of epidemic spread, in particular COVID-19 as a model of stochastic chemical kinetics**
  Various approaches to modeling the spread of epidemics, differential equations and Markov processes
- **Optimization methods for quadratic problems with large dimensionality**
  Comparison of different methods of solving high-dimensional linear regression problems
- **Intelligent Presentation Generator**
  Application for generating presentations based on text files using topic modeling

ACHIEVEMENTS

- **Fall 2025-2026:**
  - Russian Presidential Scholarship
  - Sber Scholarship Program (Top-91 students in Russia)
  - K.V. Rudakov scientific academic scholarship for research activities in the field of applied mathematics
  - Increased State Academic Scholarship for 4 year bachelor and master students at MIPT
- **Spring 2024-2025:**
  - K.V. Rudakov scientific academic scholarship for research activities in the field of applied mathematics
  - Personal scholarship for contributions to the development of numerical optimization methods
  - Increased State Academic Scholarship for 4 year bachelor and master students at MIPT
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- **Spring 2023-2024:**
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- **Fall 2023-2024:**
  - K.V. Rudakov scientific academic scholarship for research activities in the field of applied mathematics
  - Personal scholarship for contributions to the development of numerical optimization methods
- **2020-2023**:
  - Abramov scholarship for 1-3 year bachelor students with the best grades at MIPT

CERTIFICATIONS

- **Statistics for data analysis**
  Coursera, Issued Jan 2022, Credential ID KAQRGNCQJ8AH
- **Unsupervised learning**

Coursera, Issued Jan 2022, Credential ID 3CTYTEFT48FM
- **Supervised learning**
  Coursera, Issued Jan 2022, Credential ID 2ZBSN8L7EAVV
- **Mathematics and Python**
  Coursera, Issued Oct 2021, Credential ID CSTTGDM8RF2V

SKILLS

- **DL:** PyTorch, Huggingface, Accelerate, Multi-GPU training: DDP/FSDP/SP/TP, W&B, TensorBoard
- **ML:** NumPy, SciPy, Pandas, NetworkX, Scikit-learn, LightGBM, CatBoost
- **OS:** macOS, Linux, Windows
- **Misc.:** Git, Bash, LaTeX
- **Soft Skills:** responsible, organized, critical thinker, flexible, communicative, team player, patient

LANGUAGES

- Russian (Native)
- English (Advanced)

INTERESTS

- Coffee, Wine, Jazz
- Classic Guitar
- Gym, Hiking
- Traveling